

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, JavaScript, TypeScript, Java, C++, Go, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, Fastify, FastAPI, REST APIs, async services, integration testing, authentication, rate limiting

Web & Databases

React, Electron, MySQL, PostgreSQL, MongoDB, Redis, SQLite

Cloud & DevOps

Docker, Git, GitHub Actions, CI/CD, Linux, AWS, GCP, Azure

AI & Tools

PyTorch, RAG, LLM agents, OpenAI / Groq APIs, Claude Code, GitHub Copilot, Cursor

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Built and maintained **multi-tenant backend services** (Node.js, MongoDB) for **8+ clients** processing **10,000+ monthly submissions**, integrated with an external platform via REST APIs and aggregation pipelines and contributing to **\$340K+** in revenue.
- Architected a **full-stack internal tool** (React, Node.js, MongoDB) used daily by the AI team to power data-labeling processes, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved throughput by **60%**.
- Operated and hardened the services across releases: monitored production logs during feature rollouts and client integrations, shipped fixes for tenant-specific incidents, tightened authentication and rate limiting, and remediated **20+ VAPT findings**, while serving as primary technical point of contact and running knowledge transfer for new engineers.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

AgenticSearch - Provenance-First Entity Discovery Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a Python/FastAPI service running a multi-stage pipeline that plans typed retrieval, reranks results, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every value.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **full-stack desktop application** integrating camera, microphone, UI, and backend services into an on-device pipeline using MediaPipe hand tracking, gesture classification, whisper.cpp ASR, and local Text-To-Speech, designed and shipped end-to-end under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** with async APIs and queues that keep the UI responsive while streaming local video and audio with no external network dependencies.

Spark ETL Optimization - Large-Scale Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI | 1.4B-row NYC TLC dataset

- Cut end-to-end ETL runtime **25%** (17–19 min to 14.3 min) on a **1.4B-row** dataset via broadcast joins (minutes to 0.05s), AQE, skew-join handling, shuffle-partition tuning, schema-aware reads, and year/month partitioning with Snappy compression.
- Used **Spark UI** stage timings to localize shuffle and skew bottlenecks, cutting tail-latency ratio (P99/P50) from **2.14x to 1.74x** and documenting a repeatable optimization playbook.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.